

Assignment 5

Your Name Here

A microbe called *Cinderspore* is spreading between star systems and dimming stars as it goes. One research vessel, the *Meridian*, was sent out with a small crew to study it and find a way to stop it before it reaches Earth.

The *Meridian*'s crew is busy keeping the ship running and the mission alive while you are analyzing the data sent by Frank. We have a new transmission:

Mission Log 005 — from Frank

Analyst! Big day. Command finally admitted that most problems on this ship come from two things messing with each other at once, not just one thing acting alone. Now I've got three separate messes for you, and for each one it might turn out to be "it's really just Factor A," "it's really just Factor B," "it's secretly both, working together!"

Go forth. Interact responsibly.

Question 0: Data Import

0a. Call in any packages that are needed for this assignment.

0b. Import this week's datasets here.

Question 1: Cinderspore Containment Trial

Frank: "Okay, first one. We tried three different countermeasures (spray, foam, and gel) on three different surface types around the ship (hull plating, viewport glass, or insulation foam). We then measured how much of the Cinderspore we actually killed off."

This question will use the `cinderspore` dataset.

Variable	Type	Description
treatment	Categorical (3 levels)	Spray, Foam, or Gel
surface	Categorical (3 levels)	Hull Plating, Viewport Glass, or Insulation Foam
containment_pct	Continuous	Percent of Cinderspore killed off on that surface

1a. Construct the appropriate two-way ANOVA table to determine if the relationship between Cinderspore killed off and treatment type depends on surface type.

1b. Construct the appropriate hypothesis test to determine if the relationship between Cinderspore killed off and treatment type depends on surface type.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

1c. Construct a profile plot for containment_pct by treatment and surface.

Question 2: Watch Rotation Response Times

Frank: “Second one. We clocked how fast three duty tracks (Engineering, Science, and Command) respond to alerts, split out by whether they were stationed Forward or Aft at the time.”

This question will use the watch dataset.

Variable	Type	Description
duty_track	Categorical (3 levels)	Engineering, Science, or Command
zone	Categorical (2 levels)	Forward or Aft
response_time_sec	Continuous	Time to respond to a simulated alert (seconds)

2a. Construct the appropriate two-way ANOVA table to determine if the relationship between response time and duty track depends on zone.

2b. Construct the appropriate hypothesis test to determine if the relationship between response time and duty track depends on zone.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

2c. Construct an interaction plot for `response_time_sec` by `duty_track` and `zone`.

Question 3: O2 Scrubber Efficiency

Frank: “Last one, and honestly the most boring. Four scrubber models (Alpha, Beta, Gamma, Delta) were tested across three humidity zones (low, medium, high) before the next parts order goes in. I just want a comparison: which scrubber’s best and does humidity matter?”

This question will use the scrubber dataset.

Variable	Type	Description
<code>scrubber_model</code>	Categorical (4 levels)	Alpha, Beta, Gamma, or Delta
<code>humidity_zone</code>	Categorical (3 levels)	Low, Medium, or High
<code>o2_efficiency_pct</code>	Continuous	O2 output as a percent of rated capacity

3a. Construct the appropriate two-way ANOVA table to determine if the relationship between O2 output and scrubber model depends on humidity zone.

3b. Construct the appropriate hypothesis test to determine if the relationship between O2 output and scrubber model depends on humidity zone.

Hypotheses:

- $H_0 :$
- $H_1 :$

Test Statistic and p -value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

3c. Construct an interaction plot for o2_efficiency_pct by scrubber_model and humidity_zone.

3d. Construct the appropriate two-way ANOVA table to determine if the O2 output depends on either scrubber model or humidity zone.

3e. Construct the appropriate hypothesis test to determine if O2 output depends on humidity zone.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

3f. Construct the appropriate hypothesis test to determine if O2 output depends on scrubber model.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

3g. Perform posthoc testing, as appropriate, to determine which groups are different for humidity zone. Treat this as a confirmatory study. Remember to state pairwise differences.

3h. Perform posthoc testing, as appropriate, to determine which groups are different for scrubber model. Treat this as a confirmatory study. Remember to state pairwise differences.